

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION (AUTUMN 2023) — EXAMINATION QUESTION PAPER

Subject: Data Structure (CS 2001)

Time: 1.5 Hours

Semester / Programme: 3rd Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) If a 2D array $A[1..10][1..20]$ is stored in Row-Major order with base address 1000 and each element occupying 2 bytes, calculate the address of $A[5][10]$. [1]

$$\text{Address}(A[i][j]) = \text{Base} + w[N(i - L_1) + (j - L_2)]$$

1(b) Convert the following infix expression to postfix notation: $A + B * C / D$. [1]

1(c) How many null pointers are present in a Binary Tree with n nodes? [1]

$$\text{Null Pointers} = 2n - (n - 1)$$

1(d) Define a Circular Queue and write the condition for a 'Queue Full' state in its array implementation of size N (using front and rear pointers). [1]

1(e) What is the primary advantage of a Doubly Linked List (DLL) over a Singly Linked List (SLL)? [1]

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Write a pseudocode or function to insert a node at the end of a Singly Linked List. [2.5]

(b) Represent the following sparse matrix in 3-tuple format: $\{bmatrix\} 0 \ \& \ 0 \ \& \ 5 \ \backslash \ 0 \ \& \ 8 \ \& \ 0 \ \backslash \ 3 \ \& \ 0 \ \& \ 0$ [2.5]
 $\{bmatrix\}$

Q.3. (a) Evaluate the following postfix expression using a stack: 6, 2, 3, +, *, 8, 4, /, -. Show stack status at each step. [3]

(b) Explain the concept of Garbage Collection and Compaction in the context of dynamic storage management. [2]

Q.4. (a) Construct a Binary Search Tree (BST) by inserting the following elements in sequence: 50, 30, 20, 40, 70, 60, 80. Draw the final tree. [3]

(b) Write a recursive function in C/C++ to find the height of a Binary Tree. [2]
 $\text{Height}(T) = 1 + \max(\text{Height}(T.\text{left}), \text{Height}(T.\text{right}))$

Q.5. (a) A binary tree has the following traversal sequences: In-order: D, B, E, A, F, C, G Pre-order: A, B, D, E, C, F, G Construct the unique binary tree. [3]

(b) Explain how a polynomial like $P(x) = 4x^3 + 6x^2 + 10$ can be represented using a Linked List. [2]

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